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With anticipated improvements in understanding the behavior of cold-formed steel and the continuing development of new technology, this material may eventually become dated. It is anticipated that future editions of this spreadsheet will update this material as new information becomes available, but this cannot be guaranteed.

1. Cold-Formed Open Cross-Section Property Calculation

This example calculates cold-formed section properties for any open cross-section using the Coordinate Method. The centerline dimension is used. For members with holes, please see other SMath examples. Please use consistent units.

Input (provide inputs for boxed variables):

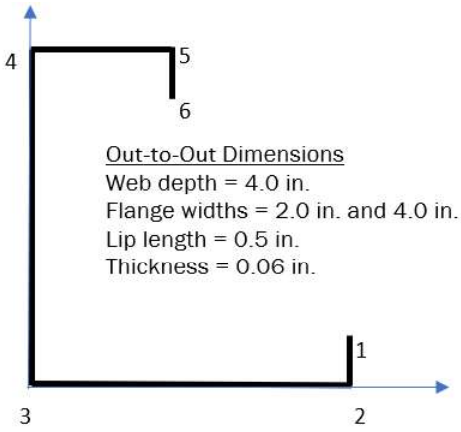
Description: $Description := \text{"Section Properties"}$

Num. of Nodes: $N := 6$
max. Nodes≡ 14

xi, yi & ti:
(fill 0 for the rest)

Note, centerline coordinates are used.

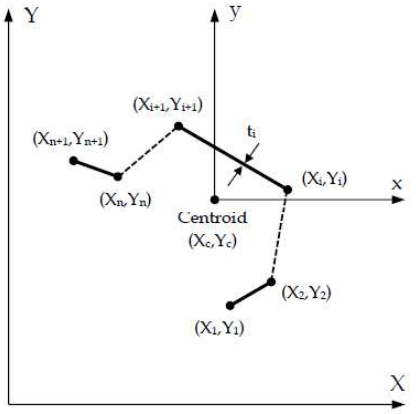
$$X := \begin{bmatrix} 4-0.03 \\ 4-0.03 \\ 0.03 \\ 0.03 \\ 2-0.03 \\ 2-0.03 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad Y := \begin{bmatrix} 0.5 \\ 0.03 \\ 0.03 \\ 4-0.03 \\ 4-0.03 \\ 3.5 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad t := \begin{bmatrix} 0.06 \\ 0.06 \\ 0.06 \\ 0.06 \\ 0.06 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$



Calculate section properties:

Element length:

$$\text{for } i \in [1..(N-1)] \qquad l_i := \sqrt{\left(X_{i+1} - X_i\right)^2 + \left(Y_{i+1} - Y_i\right)^2} \qquad l = \begin{bmatrix} 0.47 \\ 3.94 \\ 3.94 \\ 1.94 \\ 0.47 \end{bmatrix}$$



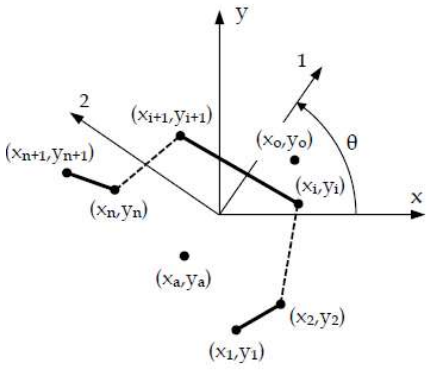
Areas:

$$\text{for } i \in [1..(N-1)] \qquad Ai_i := l_i \cdot t_i \qquad Ai = \begin{bmatrix} 0.0282 \\ 0.2364 \\ 0.2364 \\ 0.1164 \\ 0.0282 \end{bmatrix}$$

$$A := \sum_{i=1}^{N-1} Ai_i = 0.6456$$

Centroid coordinates:

$$X_c := \frac{1}{A} \cdot \sum_{i=1}^{N-1} Ai_i \cdot \frac{\left(X_i + X_{i+1}\right)}{2} = 1.1831$$



Coordinate System of Coordinate Method

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$$Y_c := \frac{1}{A} \cdot \sum_{i=1}^{N-1} Ai_i \cdot \frac{\left(Y_i + Y_{i+1}\right)}{2} = 1.6338$$

Nodal coordinates w.r.t. centroid:

$$\begin{aligned} &\text{for } i \in [1..N] \\ &\quad x_i := X_i - X_c \\ &\quad y_i := Y_i - Y_c \end{aligned} \qquad x = \begin{bmatrix} 2.7869 \\ 2.7869 \\ -1.1531 \\ -1.1531 \\ 0.7869 \\ 0.7869 \end{bmatrix} \qquad y = \begin{bmatrix} -1.1338 \\ -1.6038 \\ -1.6038 \\ 2.3362 \\ 2.3362 \\ 1.8662 \end{bmatrix}$$

Moments of inertia:

$$I_x := \frac{1}{3} \cdot \sum_{i=1}^{N-1} Ai_i \cdot \left(y_i^2 + y_i \cdot y_{i+1} + y_{i+1}^2\right) = 1.7592 \qquad r_x := \sqrt{\frac{I_x}{A}} = 1.6508$$

$$I_y := \frac{1}{3} \cdot \sum_{i=1}^{N-1} Ai_i \cdot \left(x_i^2 + x_i \cdot x_{i+1} + x_{i+1}^2\right) = 1.0548 \qquad r_y := \sqrt{\frac{I_y}{A}} = 1.2782$$

$$I_{xy} := \frac{1}{6} \cdot \sum_{i=1}^{N-1} Ai_i \cdot \left(2 \cdot x_i \cdot y_i + x_i \cdot y_{i+1} + x_{i+1} \cdot y_i + 2 \cdot x_{i+1} \cdot y_{i+1}\right) = -0.5203$$

$$I_1 := \frac{1}{2} \cdot \left(I_x + I_y\right) + \frac{1}{2} \cdot \sqrt{\left(I_x - I_y\right)^2 + 4 \cdot I_{xy}^2} = 2.0353 \qquad r_{xy} := \sqrt{\frac{\left|I_{xy}\right|}{A}} = 0.8977$$

$$I_2 := \frac{1}{2} \cdot \left(I_x + I_y\right) - \frac{1}{2} \cdot \sqrt{\left(I_x - I_y\right)^2 + 4 \cdot I_{xy}^2} = 0.7787 \qquad r_1 := \sqrt{\frac{I_1}{A}} = 1.7756$$

$$r_2 := \sqrt{\frac{I_2}{A}} = 1.0983$$

$$\theta := \text{if } \left|I_{xy}\right| \le 0.000001 \cdot \left(I_x + I_y\right) = 0.4878$$
$$\quad \text{if } I_x \ge I_y$$
$$\quad\quad 0$$
$$\quad \text{else}$$
$$\quad\quad \frac{\pi}{2}$$
$$\text{else}$$
$$\quad \frac{1}{2} \cdot \operatorname{atan}\left(\frac{-2 \cdot I_{xy}}{I_x - I_y}\right)$$

Sectorial coordinates:

$$w_{c_1} := 0$$

$$\text{for } i \in [1..(N-1)]$$
$$\quad w_{c_{i+1}} := w_{c_i} + x_i \cdot \left(y_{i+1} - y_i\right) - y_i \cdot \left(x_{i+1} - x_i\right)$$

$$w_c = \begin{bmatrix} 0 \\ -1.3098 \\ -7.6289 \\ -12.1721 \\ -16.7043 \end{bmatrix}$$

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$$\begin{bmatrix} -17.0741 \end{bmatrix}$$

$$I_{wx} := \frac{1}{6} \cdot \sum_{i=1}^{N-1} Ai_i \cdot \left(2 \cdot w_{c_i} \cdot x_i + w_{c_i} \cdot x_{i+1} + w_{c_{i+1}} \cdot x_i + 2 \cdot w_{c_{i+1}} \cdot x_{i+1} \right) = 2.1223$$

$$I_{wy} := \frac{1}{6} \cdot \sum_{i=1}^{N-1} Ai_i \cdot \left(2 \cdot w_{c_i} \cdot y_i + w_{c_i} \cdot y_{i+1} + w_{c_{i+1}} \cdot y_i + 2 \cdot w_{c_{i+1}} \cdot y_{i+1} \right) = -4.4149$$

Shear center coordinates:

$$x_o := \frac{I_y \cdot I_{wy} - I_{xy} \cdot I_{wx}}{I_x \cdot I_y - I_{xy}^2} = -2.2415 \qquad y_o := -\frac{I_x \cdot I_{wx} - I_{xy} \cdot I_{wy}}{I_x \cdot I_y - I_{xy}^2} = -0.9064$$

$$r_o := \sqrt{r_x^2 + r_y^2 + x_o^2 + y_o^2} = 3.1945$$

$$w_{o_1} := 0$$

for $i \in [1..(N-1)]$

$$w_{o_{i+1}} := w_{o_i} + \left(x_i - x_o \right) \cdot \left(y_{i+1} - y_i \right) - \left(y_i - y_o \right) \cdot \left(x_{i+1} - x_i \right) \qquad w_o = \begin{bmatrix} 0 \\ -2.3633 \\ -5.1111 \\ -0.8229 \\ -7.1136 \\ -8.5369 \end{bmatrix}$$

$$w_{o_bar} := \frac{1}{2 \cdot A} \cdot \sum_{i=1}^{N-1} Ai_i \cdot \left(w_{o_i} + w_{o_{i+1}} \right) = -3.5638$$

Normalized sectorial coordinates:

$$w_n := w_o - w_{o_bar} = \begin{bmatrix} 3.5638 \\ 1.2005 \\ -1.5473 \\ 2.7409 \\ -3.5498 \\ -4.9731 \end{bmatrix}$$

Warping constant and torsional constant:

$$C_w := \frac{1}{3} \cdot \sum_{i=1}^{N-1} Ai_i \cdot \left(w_{n_i}^2 + w_{n_i} \cdot w_{n_{i+1}} + w_{n_{i+1}}^2 \right) = 1.6952$$

$$J := \frac{1}{3} \cdot \sum_{i=1}^{N-1} Ai_i \cdot t_i^2 = 0.0008$$

Asymmetry point:

$$U_x := 0$$

for $i \in [1..(N-1)]$

$$C_1 := \left(x_i \right)^2 \cdot \left(3 \cdot y_i + y_{i+1} \right) + \left(x_{i+1} \right)^2 \cdot \left(y_i + 3 \cdot y_{i+1} \right) \qquad U_x = 0.2497$$

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$$C_2 := \left(2 \cdot x_i \cdot x_{i+1} + 3 \cdot \left(y_i\right)^2 + 3 \cdot \left(y_{i+1}\right)^2\right) \cdot \left(y_i + y_{i+1}\right)$$

$$U_x := \frac{1}{12} \cdot Ai_i \cdot \left(C_1 + C_2\right) + U_x$$

$U_y := 0$

for $i \in [1..(N-1)]$

$$C_1 := \left(y_i\right)^2 \cdot \left(3 \cdot x_i + x_{i+1}\right) + \left(y_{i+1}\right)^2 \cdot \left(x_i + 3 \cdot x_{i+1}\right)$$

$$C_2 := \left(2 \cdot y_i \cdot y_{i+1} + 3 \cdot \left(x_i\right)^2 + 3 \cdot \left(x_{i+1}\right)^2\right) \cdot \left(x_i + x_{i+1}\right)$$

$$U_y := \frac{1}{12} \cdot Ai_i \cdot \left(C_1 + C_2\right) + U_y$$

$U_y = 1.3576$

$U_w := 0$

for $i \in [1..(N-1)]$

$$C_1 := w_{n_i} \cdot \left(3 \cdot x_i^2 + 2 \cdot x_i \cdot x_{i+1} + x_{i+1}^2 + 3 \cdot y_i^2 + 2 \cdot y_i \cdot y_{i+1} + y_{i+1}^2\right)$$

$$C_2 := w_{n_{i+1}} \cdot \left(x_i^2 + 2 \cdot x_i \cdot x_{i+1} + 3 \cdot x_{i+1}^2 + y_i^2 + 2 \cdot y_i \cdot y_{i+1} + 3 \cdot \left(y_{i+1}\right)^2\right)$$

$$U_w := \frac{1}{12} \cdot Ai_i \cdot \left(C_1 + C_2\right) + U_w$$

$U_w = 0.6065$

Asymmetry point coordinates:

$$x_a := \frac{U_y \cdot I_x - U_x \cdot I_{xy}}{2 \cdot \left(I_x \cdot I_y - I_{xy}^2\right)} = 0.7945 \qquad y_a := \frac{U_x \cdot I_y - U_y \cdot I_{xy}}{2 \cdot \left(I_x \cdot I_y - I_{xy}^2\right)} = 0.3059$$

Asymmetry properties, jx and jy:

$j_x := x_a - x_o = 3.0359$

$j_y := y_a - y_o = 1.2123$

Wagner coefficients:

$\beta_x := 2 \cdot \left(y_a - y_o\right) = 2.4247$

$\beta_y := 2 \cdot \left(x_a - x_o\right) = 6.0718$

$\beta_1 := \beta_x \cdot \cos(\theta) + \beta_y \cdot \sin(\theta) = 4.9878$

$\beta_2 := \beta_y \cdot \cos(\theta) - \beta_x \cdot \sin(\theta) = 4.227$

$\beta_w := \frac{U_w}{C_w} = 0.3578$

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SUMMARY

Input:

Description: $Description = \text{"Section Properties"}$

Num. of Nodes: $N = 6$
max. Nodes≡ 14

xi, yi & ti:
(fill 0 for the rest)

$$X = \begin{bmatrix} 3.97 \\ 3.97 \\ 0.03 \\ 0.03 \\ 1.97 \\ 1.97 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad Y = \begin{bmatrix} 0.5 \\ 0.03 \\ 0.03 \\ 3.97 \\ 3.97 \\ 3.5 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad t = \begin{bmatrix} 0.06 \\ 0.06 \\ 0.06 \\ 0.06 \\ 0.06 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Section properties:

Area: $A = 0.6456$

Centroid coordinates: $X_c = 1.1831$ $Y_c = 1.6338$

Coordinates w.r.t. centroid:

$$x = \begin{bmatrix} 2.7869 \\ 2.7869 \\ -1.1531 \\ -1.1531 \\ 0.7869 \\ 0.7869 \end{bmatrix} \quad y = \begin{bmatrix} -1.1338 \\ -1.6038 \\ -1.6038 \\ 2.3362 \\ 2.3362 \\ 1.8662 \end{bmatrix}$$

Shear center coordinates: $x_o = -2.2415$ $y_o = -0.9064$

Moments of inertia: $I_x = 1.7592$ $I_y = 1.0548$ $I_{xy} = -0.5203$

$I_1 = 2.0353$ $I_2 = 0.7787$ $\theta = 0.4878$

Radii of gyrations: $r_x = 1.6508$ $r_y = 1.2782$ $r_{xy} = 0.8977$

$r_1 = 1.7756$ $r_2 = 1.0983$

Polar radius of gyration: $r_o = 3.1945$

Warping constant: $C_w = 1.6952$

Torsional constant: $J = 0.0008$

Normalized sectorial coordinates:

$$w_n = \begin{bmatrix} 3.5638 \\ 1.2005 \\ -1.5473 \\ 2.7409 \\ -3.5498 \\ -4.9731 \end{bmatrix}$$

Asymmetry point coordinates: $x_a = 0.7945$ $y_a = 0.3059$

Asymmetry properties: $j_x = 3.0359$ $j_y = 1.2123$

Wagner coefficients: $\beta_x = 2.4247$ $\beta_y = 6.0718$

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$\beta_1 = 4.9878$ $\beta_2 = 4.227$

$\beta_w = 0.3578$

Note: Bend radii can be defined using a series of short line segments, and are not considered in the above properties.

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